Ahmetcan Aksu

SOFTWARE DEVELOPER — BACKEND, FINTECH & AI

Node.js · TypeScript · C# / .NET · Rust · LLM agents & tool calling

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SUMMARY

Backend developer with 4+ years of professional experience, most recently in regulated fintech. I build the systems that move money — fund orders, bank integrations, KYC and compliance flows — in C#/.NET and Node.js/TypeScript, and I build with LLMs: tool-calling agents on Claude, OpenAI-compatible APIs and local models. My systems depth comes from designing my own programming language and VM in Rust.

HIGHLIGHTS

- **TEFAS & custodian-bank integrations** — owned end to end at Fonmap, an SPK-licensed fund platform with 100,000+ Google Play downloads.
- **100,000+ messages per hour** — real-time messaging pipeline at Qpien on WebSockets and RabbitMQ.
- **LLM agents** — AI Sprite Studio: a 16-tool agent that draws and animates pixel art, tuned for local 8B models.
- **Ellie** — a type-safe language, compiler and bytecode VM in Rust; the engine fits in 49KB; 49 GitHub stars.

EXPERIENCE

Software Developer · Fonmap Financial Technologies — Feb 2024 – Present · Istanbul, Türkiye

- **TEFAS integration:** owned the order pipeline end to end in .NET — submission, status tracking, cancellation and blockage release as fault-tolerant background jobs with distributed locking, plus the daily fund price reader; wrote 40+ of the platform's scheduled jobs.
- **Custodian-bank integration:** account opening, electronic document signing, blockage requests, money transfers and portfolio asset transfers, with request-level logging for reconciliation.
- **KYC & onboarding:** identity-document and proof-of-address upload, AML checks and sanctions screening, phone and device verification, secure session and refresh-token handling.
- **Chain transactions:** multi-step fund swaps (sell, settle, buy) that rebalance a portfolio in one request, with partial-failure and cancellation handling.
- **Regulated flows:** GSYF/GYF fund issuance and pricing, qualified-investor applications and approvals, time-stamped transaction documents, account statements and exchange reports.
- **AI in the product:** AI portfolio recommendations, portfolio outlooks and AI fund risk evaluations with plain-language explanations and disclaimers.
- **Platform & back office:** took the backend and back-office API from pre-production to go-live; notification and e-mail reporting jobs; Redis caching and bulk queries; migrated to TypeScript, then to C#/.NET (EF Core); Cloudflare R2 CDN; Elasticsearch logging.

Stack: C# / .NET, Entity Framework Core, MSSQL, Hangfire, Redis, Elasticsearch, Node.js, TypeScript, React

Full Stack Developer · Qpien — Aug 2022 – Feb 2024 · Istanbul, Türkiye

- **Real-time messaging:** WebSockets and RabbitMQ with asynchronous, parallel processing at 100,000+ messages per hour.
- **Integrations:** Instagram, Trendyol and Shopify in an all-in-one e-commerce platform; React/TypeScript front end, Node.js/GraphQL APIs on Redis and MongoDB.

Stack: React, TypeScript, Node.js, GraphQL, WebSockets, RabbitMQ, Redis, MongoDB

Full Stack Developer (Intern) · Massive Energy — Jun 2021 – Jul 2022 · Istanbul, Türkiye

- Designed and built management panels, backend architecture and databases independently; Linux deployments on AWS (Node.js, MongoDB).

Freelance Developer · Self-employed Jan 2019 – Aug 2022 · Remote

- Mobile and backend applications for clients: API design, integration and releases (React, Node.js, Flutter, Rust).

PROJECTS & OPEN SOURCE

AI Sprite Studio — Creator 2026 · TypeScript · LLM agents

- **Tool-calling agent** that edits through the same 16 undoable tools as the user; palette-indexed grids, symmetry and frame deltas keep it within the token budget of local 8B models (LM Studio).

AIStudio — AI Video Studio 2026 · TypeScript · Claude, Groq, LM Studio

- **An agent with 20+ tools** writes animated HTML video frames; deterministic export from headless Chrome under a virtual clock to ffmpeg, with ElevenLabs narration.

ProtoBoard — Creator 2026 · TypeScript · vision models, three.js

- **AI-assisted electronics prototyping:** a vision model turns a photo of a module into a to-scale part with its pads; net-aware wiring checks (shorts, over-voltage, TX/RX and I²C swaps) and a perfboard layout verified against the wiring, in 2D and 3D.

Ellie Language — Creator & Maintainer 2020 – Present · Rust · 49 GitHub stars
github.com/behemehal/Ellie-Language

- **Type-safe language** for embedded (ARM) and sandboxed (WebAssembly) targets: parser, compiler and bytecode VM built from scratch, including memory management and garbage collection.

Open Source & Products — Behemehal 100+ repositories · Rust · TypeScript · Dart
github.com/behemehal · github.com/ahmetcanaksu

- **UtilStation** (utilstation.com, 85+ developer tools); Rust crates Menemen, Rust-NMEA, SafeEn and Wole on crates.io; CHIP-8 and 6502 emulators; Flutter apps.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, C#, Rust, Dart, C, C++, Java, SQL
AI & LLMs	LLM agents & tool calling, Claude API, OpenAI-compatible APIs (Groq, LM Studio), local LLMs (Qwen3-VL, Gemma), vision models
Backend	Node.js, Deno, Express, GraphQL (Apollo), REST, WebSockets, .NET, Entity Framework Core, Hangfire, RabbitMQ
Frontend	React, Redux, Svelte, Electron, Flutter, HTML/CSS, WebAssembly
Databases	MSSQL, PostgreSQL, MongoDB, Redis, Elasticsearch, MinIO
Infra & Tools	Docker, Linux, AWS, Cloudflare R2, Git, Puppeteer, CI/CD
Systems & Embedded	Algorithm design, compiler and bytecode VM design, ARM, ESP32/ESP8266, Arduino, Raspberry Pi, TCP/UDP, FFI
Spoken	Turkish — Native; English — Professional — C2 (EF SET certified)

EDUCATION

BSc, Management Information Systems
Anadolu University · 2024 – Present

Associate Degree, Computer Programming
Işık University · 2022 · GPA 3.53 / 4.00 · C, C++, C#, Java, JavaScript, MSSQL, HTML

CERTIFICATES

EF SET English Certificate — C2; Turkish Capital Markets Association; Mobile App Development with Google Flutter.